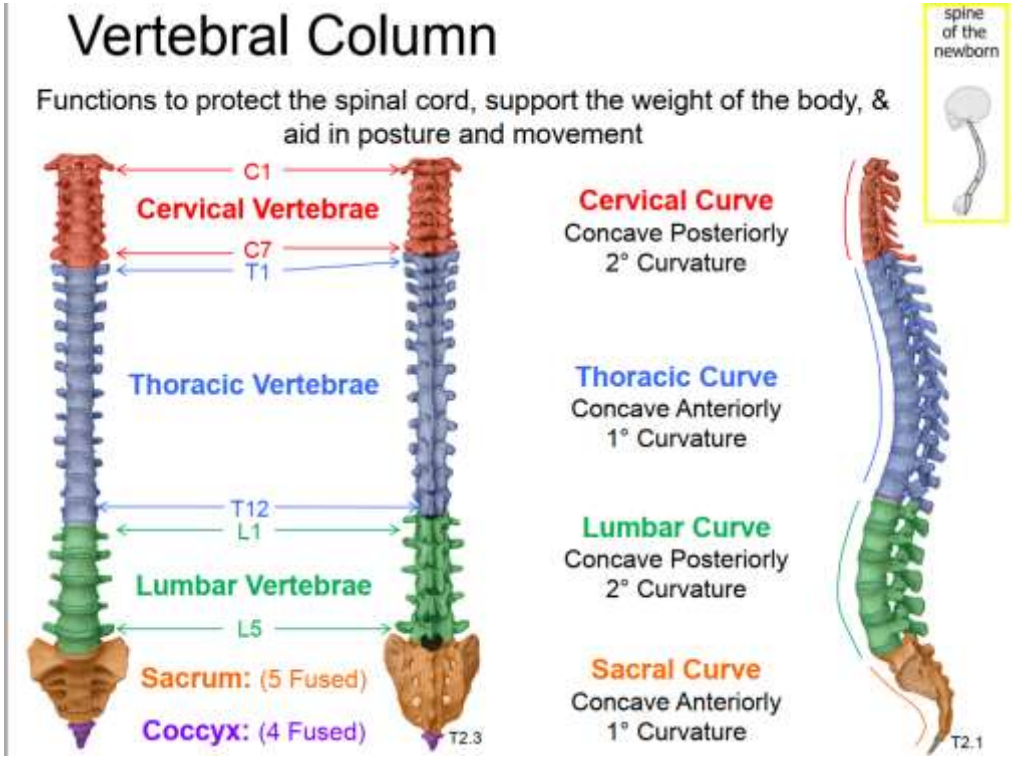
Week #8: [Mechanical Back & Neck Pain]

(Non-hyperlinked = Dr. Caitlin Cassidy Module)

• What's the Big Deal? • Mechanical Axial Pain
• **Anatomy of the Bones, Joints, and Ligaments of the Back**
• Disc
• Facet
• **Anatomy of the Back Muscles**
• Non-Mechanical Axial Pain
  - Neurology edition
• Problems to the side – radiculopathy
• Problems to the centre
• Lumbar (RED FLAGS)
• Cervical (RED FLAGS)
• "Everything else" – RED FLAGS
  - Inflammatory
  - Fractures
  - Visceral
  - Infection
  - Neoplasm

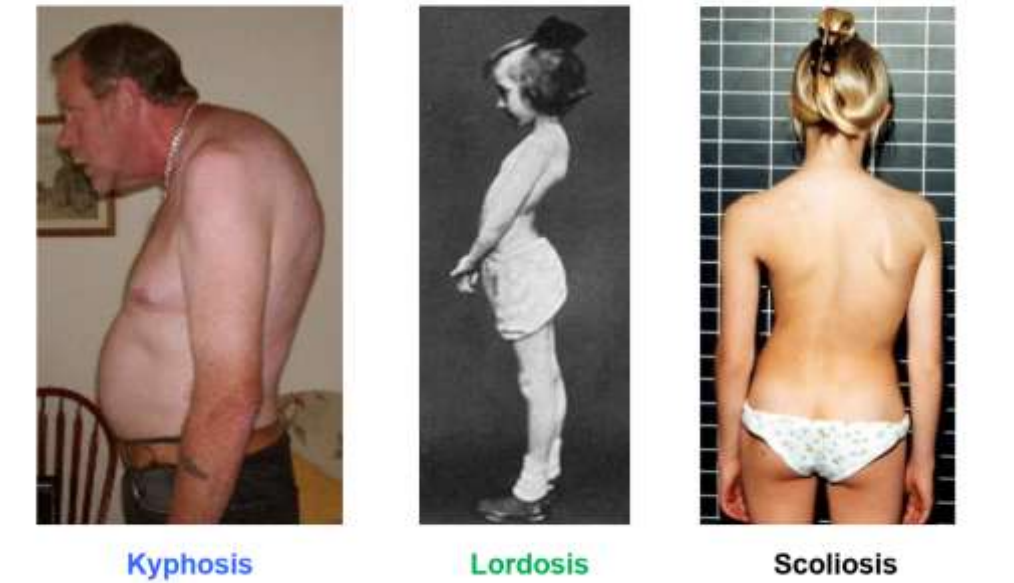
Anatomy of the Bones, Joints, and Ligaments of the Back



The vertebral column extends from the base of the skull to the coccyx and functions to protect the spinal cord, support the weight of the body, and aid in posture and movement

Consists of 33 vertebrae:

1. **7 Cervical** vertebrae
2. **12 Thoracic** vertebrae
3. **5 Lumbar** vertebrae
4. **5 Sacral** vertebrae – fused to form the **sacrum**
5. **4 Coccygeal** vertebrae – fused to form the **coccyx**

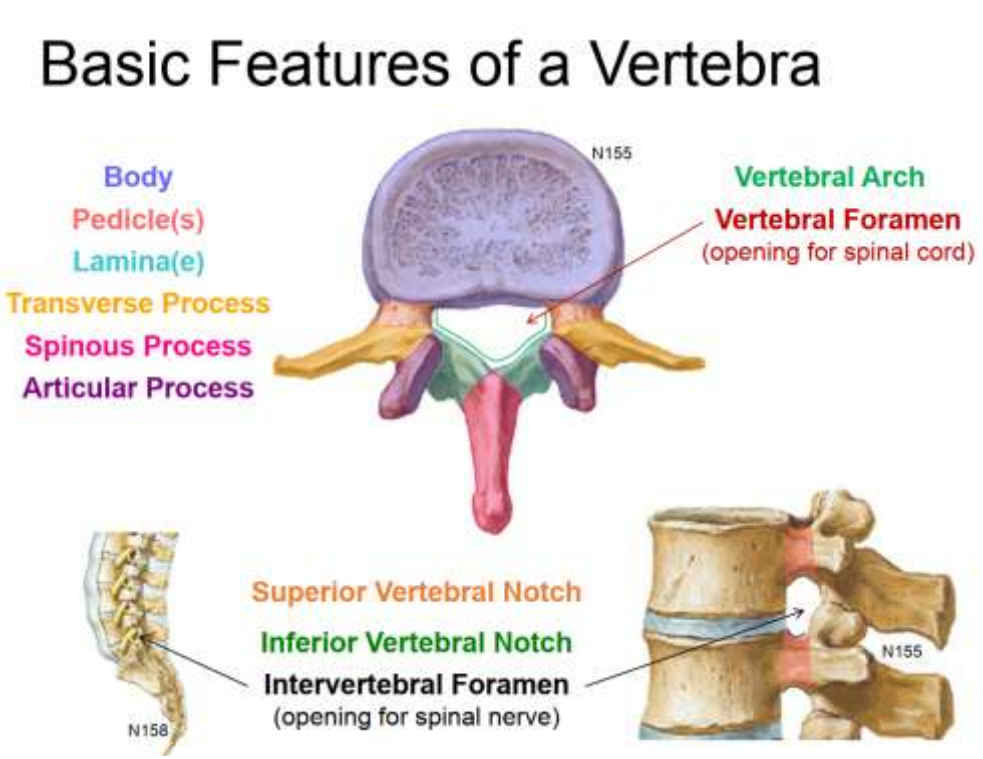


Curvatures of the Vertebral Column:

1. Normal Curves:
    a. **Thoracic** and **sacral curvatures** are concave **anteriorly**
    b. **Cervical** and **lumbar curvatures** are concave **posteriorly**
2. Abnormal Curvatures:
    a. **Kyphosis** – exaggerated *thoracic curvature*, typically the result of erosion or fracture of the anterior part of the vertebrae, due to **osteoporosis** or other aging factors
    b. **Lordosis** – exaggerated *lumbar curvature*, characterized by anterior rotation of the pelvis. Often cause by weakened trunk musculature and seen during pregnancy
    c. **Scoliosis** – increased *lateral curvature* of the spine with rotation of the vertebrae. May be due to weakness of back muscles, poor posture, congenital defects in vertebrae development, and inequality in the length of the lower limbs

Features of a Vertebra:

1. **Vertebral Body** – large anterior portion that provides strength and permits transmission of body weight. Vertebral bodies become larger as they progress inferiorly in order to support increasingly greater body weight
2. **Vertebral Arch** – posterior portion of vertebra, composed of fused pedicles

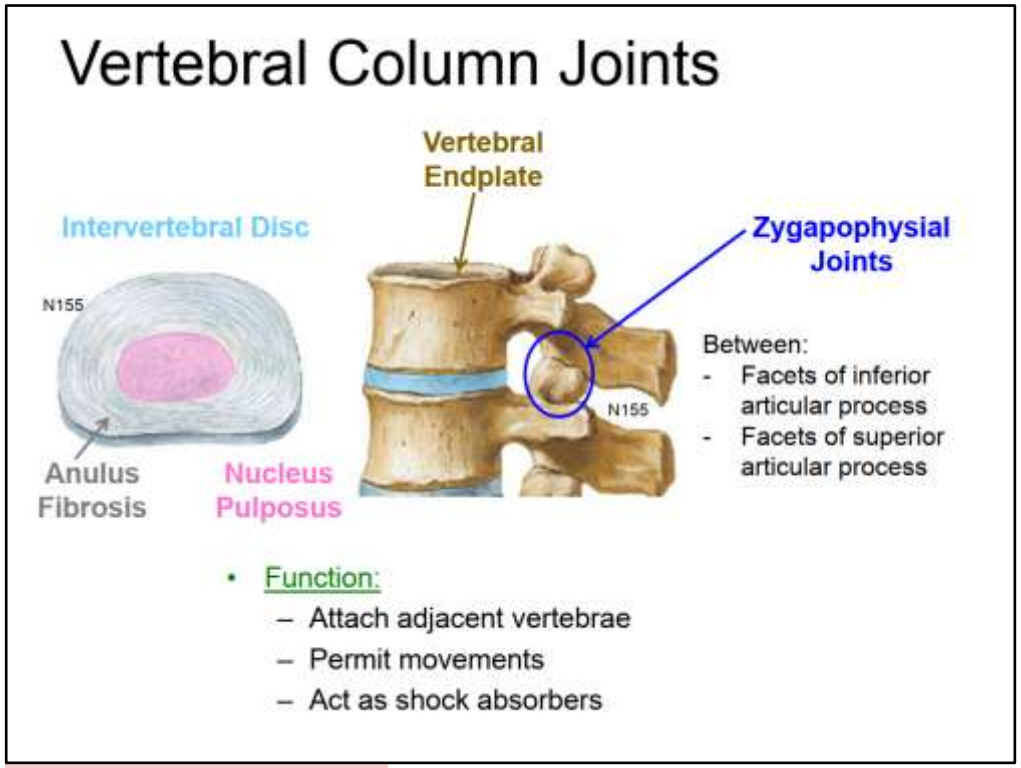


and lamina

   a. ***Pedicles*** – short, stout processes that join the vertebral arch to the vertebral body. Each pedicle has superior and inferior vertebral notches that form the intervertebral foramen
   b. ***Lamina*** – 2 broad, flat plates that close the vertebral arch posteriorly

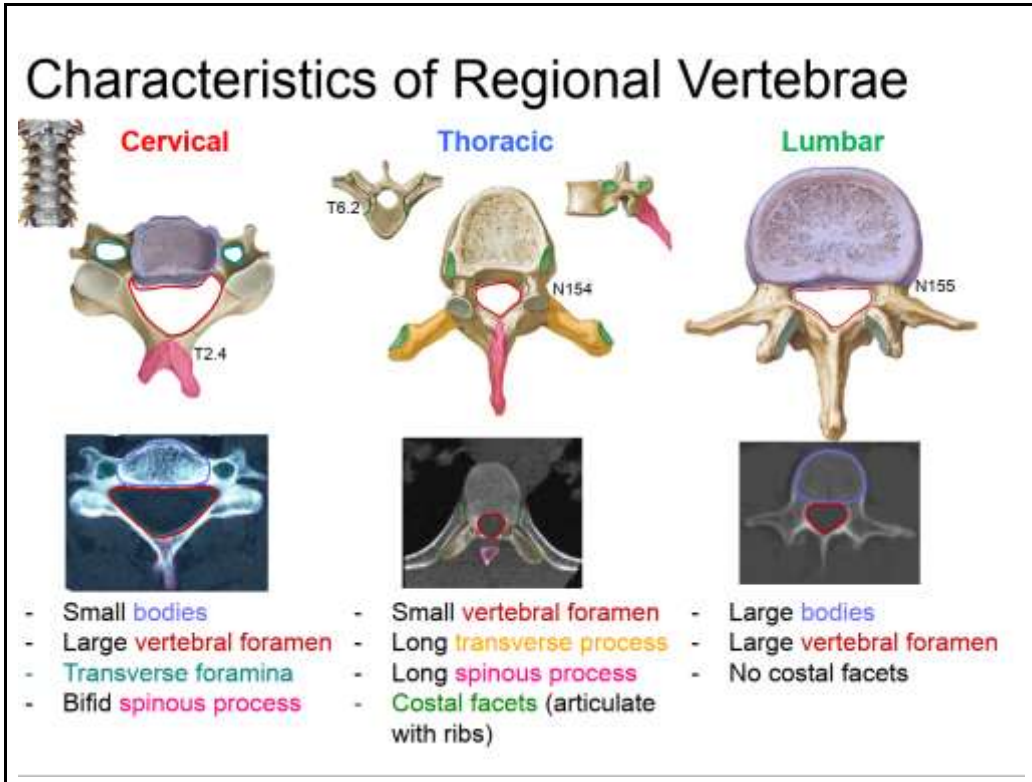
3. **Vertebral Foramen** – opening for spinal cord; surrounded by the posterior wall of the vertebral body and the vertebral arch
4. **Spinous Process** – projects posteriorly from the vertebral arch at the junction of the two laminae; attachment site for muscles and ligaments
5. **Transverse Processes** – two processes that project posterolaterally from junction of pedicles and laminae; attachment site for muscles and ligaments
6. **Articular processes** – four processes, 2 superior and 2 inferior, that articulate with adjacent vertebrae forming zygapophysial joints. Located at the junction of the pedicles and laminae

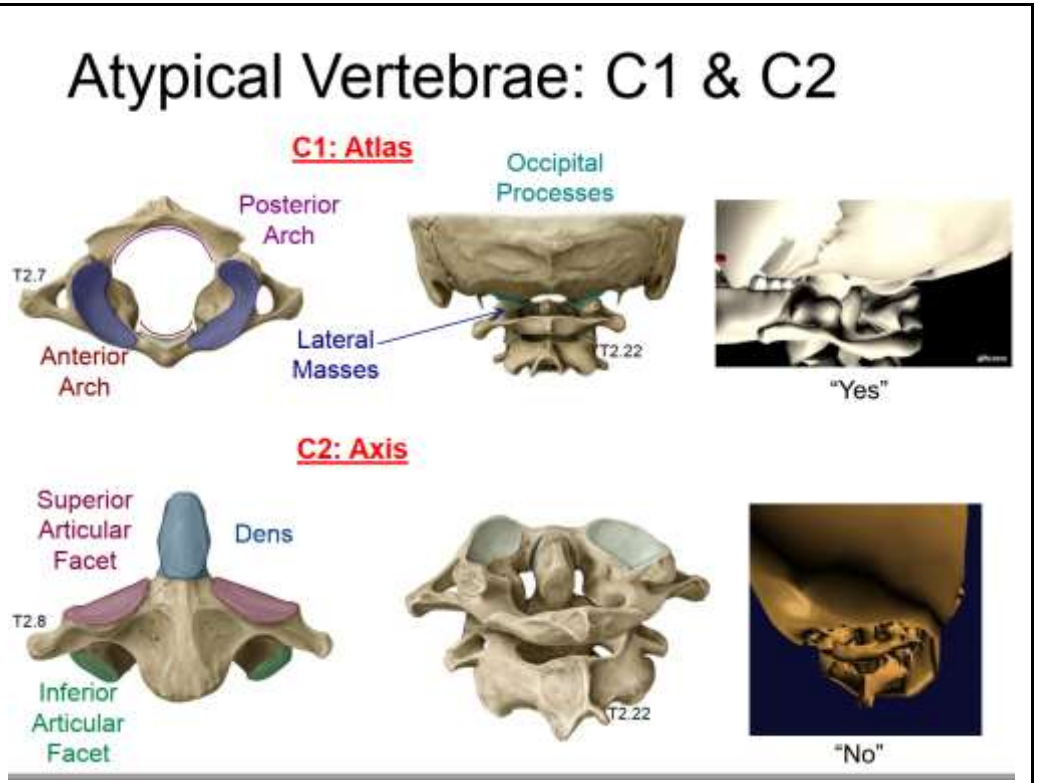
Clinical Note: a **laminectomy** is the surgical excision of one or more spinous processes and their supporting laminae in a particular region of the vertebral column. Laminectomies provide access to the vertebral canal to relieve pressure on the spinal cord or nerve roots, commonly caused by a tumor, herniated IV disc, or boy hypertrophy (excess growth).



Joints of the Vertebral Column:

1. Each vertebra (except C1 that attaches to the skull and L5 that attaches to the sacrum) attaches to the superiorly and inferiorly adjacent vertebra at the bodies of the vertebrae via **intervertebral discs** and at the **zygapophysial joints**. The intervertebral discs interface with the vertebral bodies at a transitional area known as the vertebral end plate – The vertebral end plate is comprised of two layers a cartilaginous layer that fuses with the disc and a thin layer of porous bone attached to the vertebral body.
2. **Intervertebral (IV) Discs** are located between the vertebral bodies. They are strong attachments between the vertebrae, they permit movement, and serve as shock absorbers
    a. **Components:**
        i. **Anulus Fibrosis** – outer portion, consisting of concentric layers of fibrocartilage
        ii. **Nucleus Pulposus** – inner portion, consisting of a gelatinous matrix with high water content
    a. **Components (continued)**
        a. Thickness of IV discs varies with vertebral column region (cervical and lumbar regions are thicker) iii. IV discs account for ~25% of the vertebral column height
        b. There is no IV disc between C1 and C2
        c. **Clinical Note: Disc herniation**, or protrusion of the nucleus pulposus into or through the annulus fibrosus, is a well-recognized cause of back pain. Herniated material may directly compress nerve roots or the spinal cord itself, resulting in pain and/or neurological deficits. Most posterolateral herniations occur between L4//L5 or L5/S1, typically affecting the nerve root immediately inferior to it (i.e., the L4 disc impinges on the L5 nerve root).
2. **Zygapophysial Joints** are plane synovial joints between the superior and inferior articular processes of adjacent vertebrae.
3. **Craniovertebral joints:**
    a. **Atlanto-occipital joints** – formed between the atlas (C1) and the occipital condyles
        i. Permit flexion and extension (nodding) of the head
    b. **Atlanto-axial joints** – formed between the atlas (C1) and axis (C2)
        i. Permit the head to turn from side to side (saying 'no') via pivoting of C1 around the dens of C2
    c. **Costovertebral joints** – a typical rib articulates posteriorly with the vertebral column at two joints:
        i. Joints at the heads of the ribs:
            1. The head of the rib articulates with the superior costal facet of the corresponding (same- numbered) vertebra
            2. The inferior costal facet of the vertebra superior to it
            3. The adjacent IV disc uniting the two vertebrae
    d. **Costotransverse joints** - the tubercle of the rib articulates with the articular facet on the transverse process of the corresponding (same-numbered) rib





Regional Vertebrae Differences:

1. Cervical:

a. ***Vertebral body:*** small
b. ***Vertebral foramen:*** large; especially C3-C7 due to the enlargement of the spinal cord
c. ***Transverse processes:*** short, with transverse foramina (openings) in all cervical vertebrae
   i. Vertebral artery ascends to the skull through the transverse foramen from C6-C1
   ii. Only the vertebral vein passes through the C7 transverse foramina
d. ***Articular processes:*** superior articular facets directed superoposteriorly; inferior facets directed inferoanteriorly
e. ***Spinous processes:*** short and bifid, with exception of C7, which has an exaggerated spinous process called the vertebra prominens (easily palpated when neck is flexed)
f. ***Movements:*** flexion and extension
g. C1 and C2 are atypical vertebrae:
   i. Atlas (C1) – ring shaped; consisting of two **lateral masses** (with superior and inferior articular processes) that are connected by an **anterior arch** (replaces body) and a **posterior arch** (instead of lamina). On its cranial surface is a groove for the vertebral artery
   ii. Axis (C2) – its distinguishing feature is the **dens**, which extends superiorly from the C2 body and forms a pivot joint with the articular facet on the posterior surface of the anterior arch of C1

2. Thoracic:

a. ***Vertebral body:*** heart-shaped with **costal facets** for articulation with the ribs via **costovertebral joints** and **costotransverse joints**
b. ***Vertebral foramen:*** circular and smaller than those in the cervical and lumbar regions
c. ***Transverse processes:*** long; transverse costal facets for articulation with the tubercle of rib
d. ***Articular processes:*** superior articular facets are directed posteriorly and slightly laterally; inferior articular facets are directed anteriorly and slightly medially.

This orientation prevents excessive flexion and extension

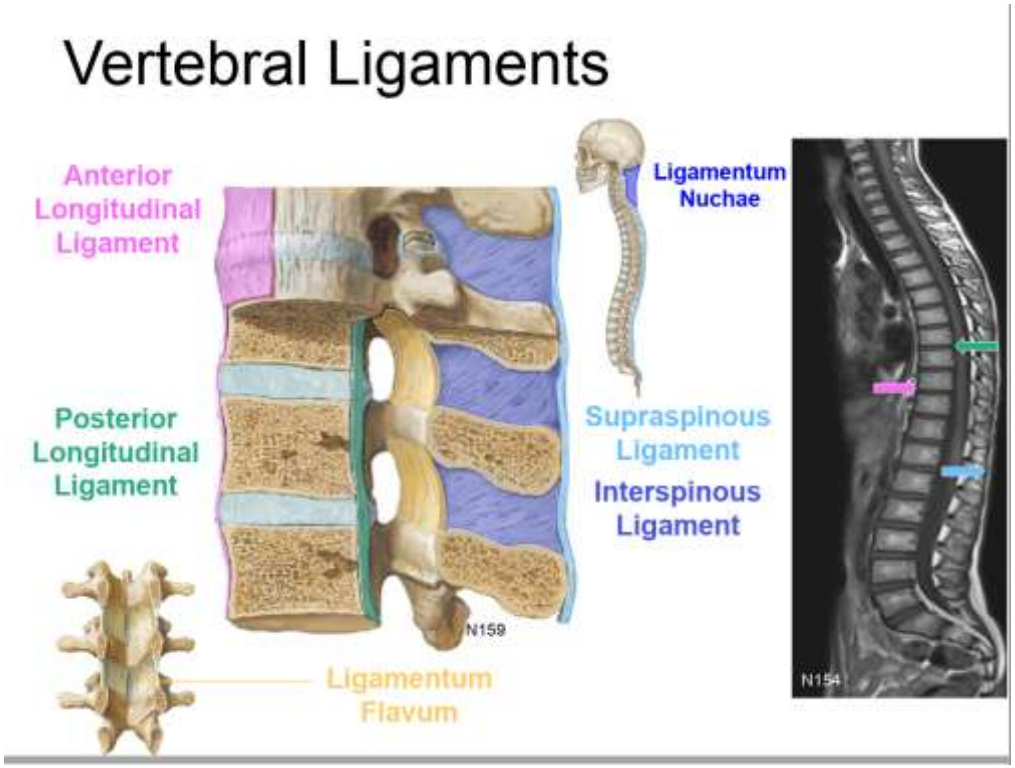
e. ***Spinous processes:*** long; slopes inferiorly and overlaps vertebral arch inferior to it
f. ***Movements:*** rotation and some lateral flexion

3. Lumbar:

a. ***Vertebral body:*** larger, strong vertebrae with short, heavy pedicles
b. ***Vertebral foramen:*** triangular-shaped (larger than thoracic vertebral foramina)
c. ***Transverse processes:*** long and slender transverse processes, with no costal facets
d. ***Articular processes:*** superior articular facets are directed posteromedially; inferior articular facets are directed anterolaterally
e. ***Spinous processes:*** short and stout
f. ***Movements:*** flexion, extension, some lateral bending, and limited rotation

G. Sacrum:

a. Consists of 5 fused vertebrae that provide strength and stability to the pelvis and transmits body weight to the ilium through the **sacroiliac joints**
b. Anterior (pelvic) surface is smooth and concave, while the posterior surface is rough and convex
c. **Sacral Foramina** – paired openings on the anterior and posterior sides to transmit the sacral nerves
d. **Sacral Hiatus** – inverted U-shaped opening that leads into the sacral canal (inferior end of the vertebral canal); this is one side for a caudal epidural anesthesia



Vertebral Column Ligaments:

a. **Anterior Longitudinal Ligament** – strong, broad fibrous band that covers the anterolateral aspect of the vertebral bodies and IV discs
    i. Runs from the occipital bone to the sacrum
    ii. Helps resist *hyperextension* of the vertebral column
b. **Posterior Longitudinal Ligament** – narrow ligament that runs within the vertebral canal along the posterior aspect of the vertebral bodies
    i. Helps resist hyperflexion of the vertebral column
c. **Ligamentum Flava** – broad, pale yellow fibrous tissue joining adjacent vertebral laminae
d. **Supraspinous Ligaments** – connects the tips of the spinous processes from C7 to the sacrum
    i. Above C7, these fibres merge with the ***nuchal ligament***, which is an extremely strong band stretching from the external occipital protuberance to the spinous processes of the cervical vertebrae; the ***nuchal ligament*** extends from the spinous processes to the skin filling the cervical curvature; serves as an attachment point for the muscles of the neck and upper back
e. **Interspinous Ligaments** – thin ligament that connects adjacent spinous processes
f. **Intertransverse Ligaments** – connect adjacent transverse processes

Knowledge Check

A





Large vertebral body
Lumbar vertebra

B

Cervical vertebra





Small vertebral body
Transverse process
Short spinous process

C

Thoracic



heart shaped



long spinous process

Knowledge Check

C7 Prominens (B) C7 prom

Facet Joints (C) - facet joints b/t vertebra

Intervertebral Disc (E)

Spinous Process (D) - sticks out w/ tab

Vertebral Body (A) of C5



Knowledge Check

- MVR space on lumbar vertebra

Facet Joints (D)

Inferior Vertebral Notch (C) - inferior asp of pedicle

Intervertebral Disc (G) - larger space

Intervertebral Foramen

Sacrum (H) - L5 + sacrum promontory

Spinous Process (A) thick up

Superior Vertebral Notch (E)

Vertebral Body (B) - sig larger



PART II: Anatomy of the Back Muscles

Back Musculature

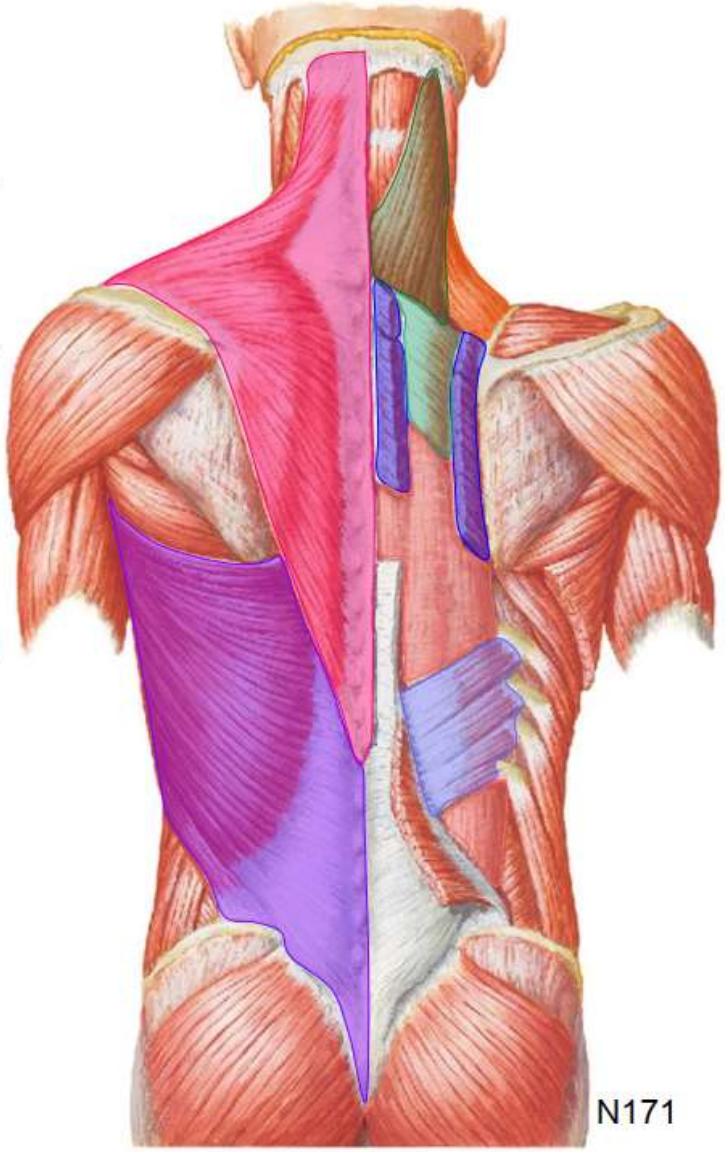
Superficial Musculature

Trapezius

Latissimus Dorsi

Rhomboids

Levator Scapulae



Deep Musculature

Splenius

Erector Spinae

Transversospinalis

Intermediate Musculature

Serratus Posterior Superior

Serratus Posterior Inferior

Trapezius



Origin:
Superior Nuchal Line
External Occipital Protuberance
Ligamentum Nuchae
Spinous Processes C7-T12

Insertion:
Lateral ⅓ of Clavicle
Acromion Process
Spine of Scapula

- Function:
  - Elevates, depresses and retracts scapula
  - Rotates glenoid cavity superiorly
- Innervation:
  - Spinal Accessory Nerve (CN XI)



Latissimus Dorsi

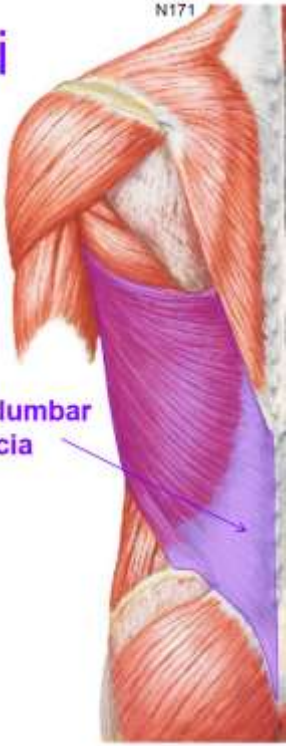


Insertion:
Intertubercular Groove

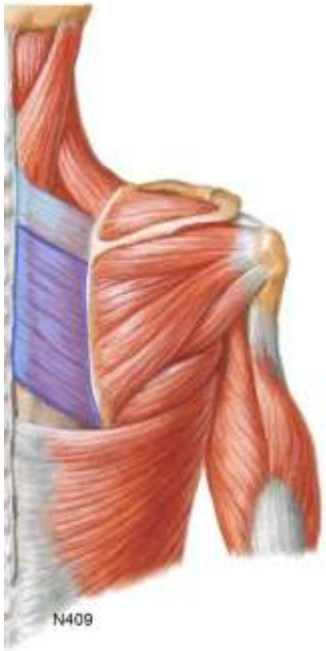


Origin:
Inferior 3-4 Ribs
Spinous Processes of T7-12
Thoracolumbar Fascia
Iliac Crest

- Function:
  - Extends, adducts, and medially rotates humerus
- Innervation:
  - Thoracodorsal Nerve



Rhomboids



Origin:
Minor: Ligamentum Nuchae
C7-T1 Spinous Processes
Major: T2-5 Spinous Processes

Insertion:
Minor: Medial border at root of the spine
Major: Medial border inferior to root of the spine



- Function:
  - Retract scapula
  - Rotate to depress glenoid cavity
  - Fix Scapula to thoracic wall
- Innervation:
  - Dorsal Scapular Nerve

Levator Scapulae



Origin:
C1-4 Transverse Processes

Insertion:
Medial border superior to root of the spine



- Function:
  - Elevates scapulae
  - Rotates to depress glenoid cavity
- Innervation:
  - Dorsal Scapular Nerve
  - C3-4 Spinal Nerves

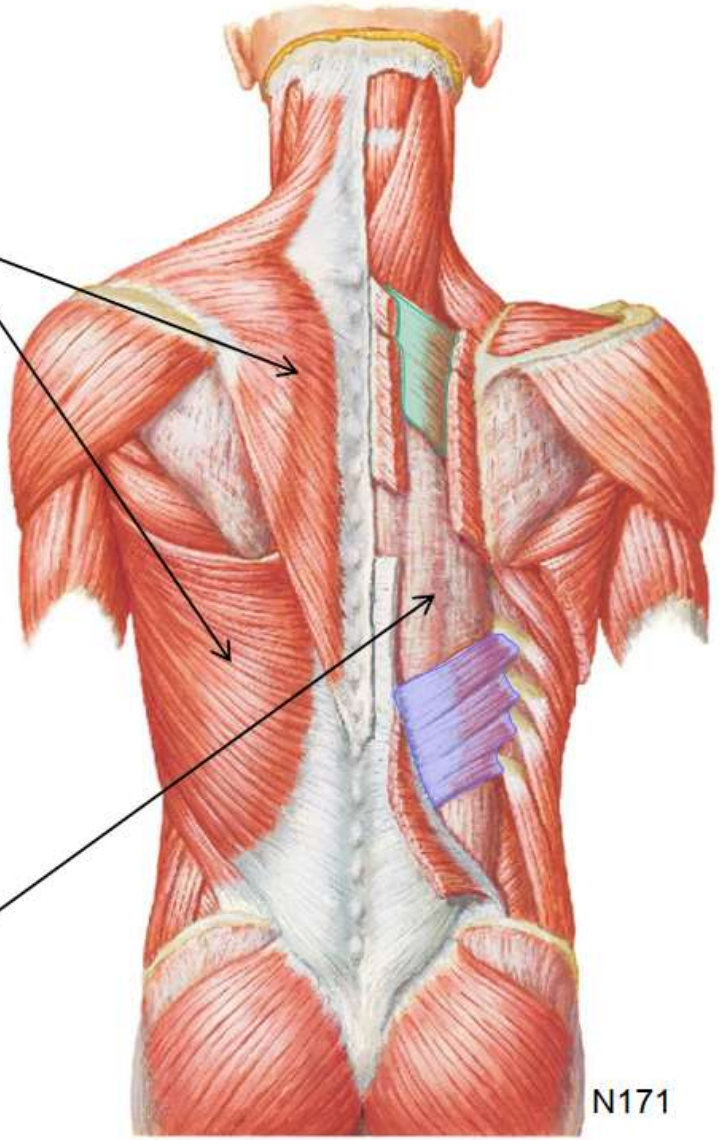
Superficial (Extrinsic) Muscles of the Back

Muscle	Description	Origin	Insertion	Main Action(s)	Innervation
Trapezius	Large trapezoidal-shaped muscle; covers neck and upper trunk	Superior nuchal line; external occipital protuberance; ligamentum nuchae, spinous processes of C7-T12	Lateral ⅓ of clavicle, acromion, and spine of scapula	Elevates, depresses, and retracts scapula; rotates glenoid cavity superiorly	Spinal Accessory Nerve
Latissimus Dorsi	Large, flat, fan-shaped muscle; Origin is dense and aponeurotic	Spinous processes of T7-12, **thoracolumbar fascia**, iliac crest, and inferior 3-4 ribs	Intertubercular groove of humerus	Extends, adducts, and medially rotates humerus	Thoracodorsal Nerve
Levator Scapulae	Relatively thin, strap like muscle	Transverse processes of C1-C4	Medial border of scapula superior to root of spine	Elevates scapula and rotates to depress glenoid cavity	Dorsal Scapular Nerve & C3-4 Spinal Nerves
Rhomboid Major	Deep to trapezius; major is inferior to minor; they are often fused	Spinous processes of T2-5	Medial border of scapula inferior to root of spine	Retract scapula and rotate to depress glenoid cavity; fix scapula to thoracic wall	Dorsal Scapular Nerve
Rhomboid Minor	Deep to trapezius; minor is superior to major; they are often fused	Nuchal ligament; spinous processes of C7-T1	Root of the spine of the scapula	Retract scapula and rotate to depress glenoid cavity; fix scapula to thoracic wall	Dorsal Scapular Nerve

Intermediate Musculature

Superficial Musculature

Deep Musculature



Intermediate Musculature

Serratus Posterior Superior

Serratus Posterior inferior

Intermediate (Extrinsic) Muscles of the Back

Muscle	Description	Origin	Insertion	Main Action(s)	Innervation
Serratus Posterior Superior	Thin, saw-toothed muscles located deep to the rhomboids	Ligamentum nuchae; spinous processes of C7-T3	Superior borders of ribs 2-4	Elevates ribs; proprioception	2nd-5th Intercostal Nerves
Serratus Posterior Inferior	Thin, saw-toothed muscles located deep to latissimus dorsi	Spinous processes of T11-L2	Inferior borders of ribs 8-12	Depresses ribs; proprioception	9th-12th Intercostal Nerves

Deep Back Musculature

Superficial Musculature
- Trapezius
- Latissimus Dorsi
- Rhomboids
- Levator Scapulae



Deep Musculature
- Splenius
- Erector Spinae
- Transversospinalis

Intermediate Musculature
- Serratus Posterior Superior
- Serratus Posterior Inferior

Erector Spinae

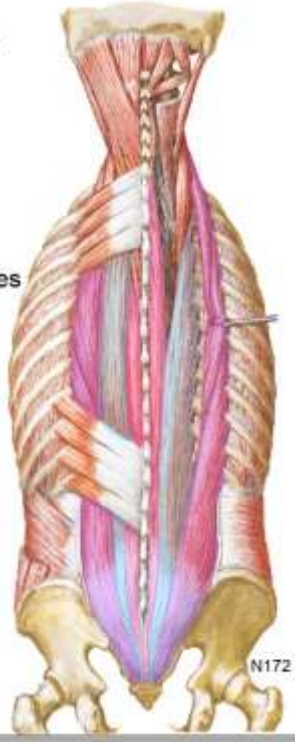


- Iliocostalis
- Longissimus
- Spinalis

Insertion:
- Iliocostalis: rib & superior transverse processes
- Longissimus: ribs & superior transverse processes
- Spinalis: superior spinous processes

Origin:
Broad tendon from sacrum, iliac crests, & inferior spinous processes

- Function:
  - Acting Bilaterally: Extend vertebral column & head
    Eccentrically control trunk flexion
  - Acting Unilaterally: Laterally flex vertebral column
- Innervation:
  - Dorsal Rami of Spinal Nerves

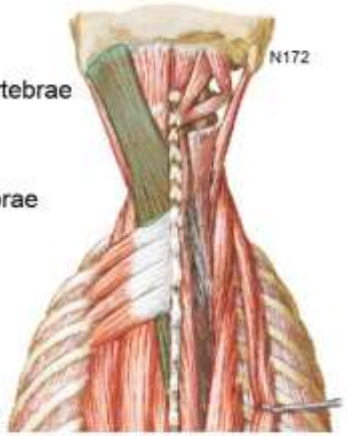


Splenius Muscle Group

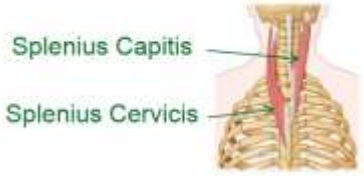


Insertion:
Occipital Bone
Transverse Processes of Upper Cervical Vertebrae

Origin:
Ligamentum Nuchae
Spinous Processes of Upper Cervical Vertebrae



- Function:
  - Acting Bilaterally: Extend head & neck
  - Acting Unilaterally: Laterally flex & rotate head & neck
- Innervation:
  - Dorsal Rami of Spinal Nerves



Transversospinalis Muscle Group



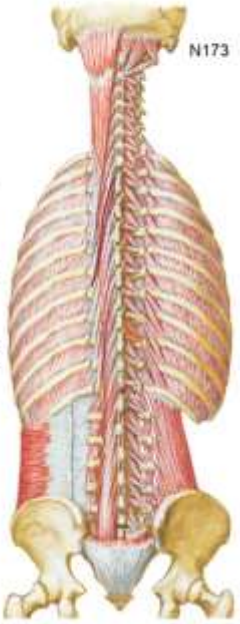
Insertion:
Spinous Process

Origin:
Transverse Process

Semispinalis
Spans 4-6 segments

Multifidis
Spans 2-4 segments

Rotatores
Spans 1-2 segments



- Function:
  - Contralateral rotation
  - Stabilization
  - Proprioception
- Innervation:
  - Dorsal Rami of Spinal Nerves

Deep (Intrinsic) Muscles of the Back

1. Maintain posture and control movements of the vertebral column
2. Innervated by posterior rami of spinal nerves; these posterior rami ultimately become cutaneous nerves that provide sensation to the skin of the back and neck.

Muscle	Description	Origin	Insertion	Main Action(s)	Innervation
<u>Erector Spinae</u> **Iliocostalis** **Longissimus** **Spinalis**	**Iliocostalis:** Most lateral **Longissimus:** intermediate muscle; usually widest **Spinalis:** most medial	Arises via a broad tendon from the sacrum, iliac crests, lumbar and thoracic spinous processes before dividing into three vertical columns of muscle	**Iliocostalis:** rib and cervical transverse processes **Longissimus:** ribs and transverse processes in the thoracic and cervical regions, and cranium **Spinalis:** spinous processes in upper thoracic region and cranium	*Acting bilaterally*: extend vertebral column and head; eccentrically control trunk flexion *Acting unilaterally*: laterally flex vertebral column	Dorsal Rami of Spinal Nerves
Splenius	Composed of two parts: **cervicis** and **capitis**; both ascend in a medial to lateral direction	Ligamentum nuchae and spinous processes of upper thoracic vertebrae	Occipital bone and transverse processes of upper cervical vertebrae	*Acting bilaterally*: extends head and neck *Acting unilaterally*: laterally flexes and rotates head and neck	Dorsal Rami of Spinal Nerves
Transverso-spinalis	Three muscles of different lengths are a part of this muscle group: Semispinalis, Multifidus, Rotatores; span 1-6 vertebral levels	Transverse Processes	Spinous Processes	Contralateral rotation and stabilize vertebrae and proprioception	Dorsal Rami of Spinal Nerves

Knowledge Check

Erector Spinae (D) – extend / contract as you doing controlled flexion of trunk

Dorsal Ram (E)

Latissimus Dorsi (B)

Trapezius (A) 3 fibers: inferior, middle, superior

Rhomboids (C)

